

Reg. No. :

Name :



Second Semester M.C.A. Degree Examination, December 2022

MCA 50 C 24: DESIGN AND ANALYSIS OF ALGORITHMS

(2020 Scheme)

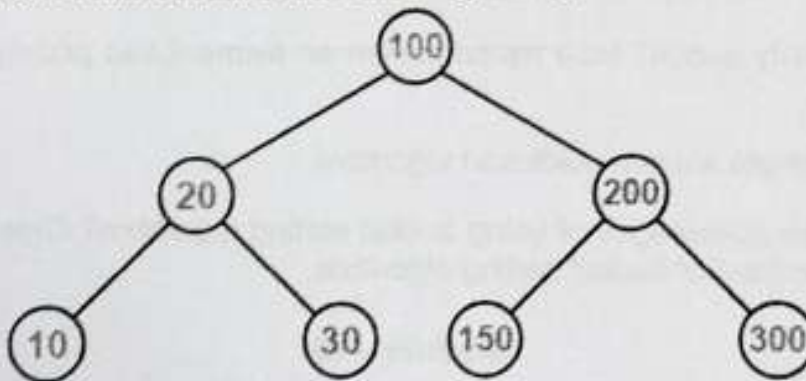
Time : 3 Hours

Max. Marks : 100

SECTION – A

Answer **all** questions. **Each** question carries **4** marks.

1. How is maximum Subarray problem solved using Divide and conquer method?
2. What do you mean randomized algorithm? What are the two main types of randomized algorithms?
3. Create a max heap using the following elements. Show each step in construction of heap. 3, 1, 6, 5, 2, 4.
4. What do you mean by the i^{th} order statistic of a set of n elements? Explain.
5. Find the preorder traversal of the following binary search tree.



P.T.O.

6. What is linear probing?
7. Explain Push Relabel Algorithm.
8. What is maximum matching in bipartite graph?
9. What is the time complexity of Rabin-Karp algorithm?
10. What is the difference between NP hard and NP complete problem?

(10 × 4 = 40 Marks)

SECTION – B

Answer **any two** questions from each module. **Each** question carries **6 marks**.

Module – I

11. Explain algorithm for Insertion sort and analyze its complexity.
12. Solve the following recurrence relation using Master's theorem:
 $T(n) = 3T(n/2) + n^2$.
13. Explain in detail the hiring problem.

Module – II

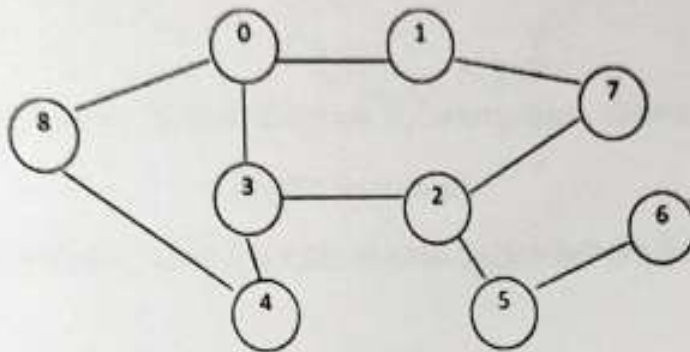
14. What is priority queue? How we can insert an element into priority queues using max heaps.
15. With an example explain quick sort algorithm.
16. What are the advantages of using bucket sorting algorithm? Give an example and show the working of bucket sorting algorithm.

Module – III

17. Explain quadratic probing. What is the disadvantage of quadratic probing.
18. With an example explain insertion operation in a red black tree.
19. Explain Matrix chain multiplication algorithm with an example.

Module – IV

- 1. Explain in detail about B-trees.
- 2. Find the DFS traversal of the graph given below (start at 0 vertex). Explain the algorithm of DFS traversal.



- 22. Explain Prim's algorithm with an example.

Module – V

- 23. Explain in detail about multithreaded merge sort.
- 24. How does Knuth Morris Pratt algorithm work?
- 25. With an example explain how string matching is done using finite automata.

(10 × 6 = 60 Marks)